

Lecture 9, Feb 9, 2026.

* Indeterminate Products.

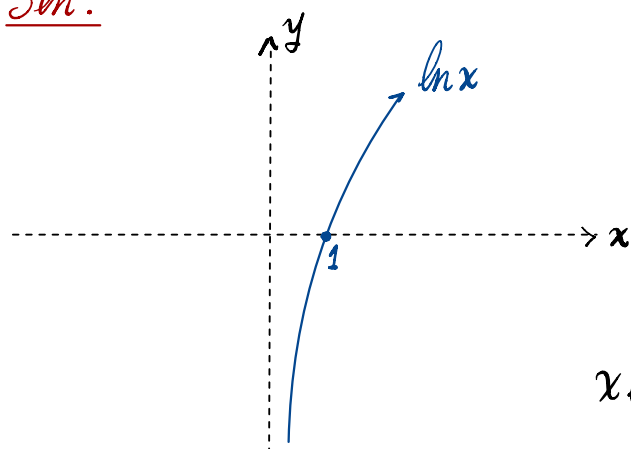
Let consider h, g such that $f(x) = h(x)g(x)$ Notice that.

$$f(x) = \frac{g(x)}{\frac{1}{h(x)}} = \frac{h(x)}{\frac{1}{g(x)}}$$

Indeterminate form of type $0 \cdot \infty$.

Example. Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Sln.



Notice that $\ln x \rightarrow -\infty$ as $x \rightarrow 0^+$

and $x \rightarrow 0$ as $x \rightarrow 0^+$

then we have $0 \cdot \infty$. then let

$$x = \frac{1}{1/x} \quad \text{then}$$

$$x \ln x = \frac{\ln x}{1/x}$$

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}; \quad \begin{array}{l} \ln x \rightarrow -\infty \text{ as } x \rightarrow 0^+ \\ 1/x \rightarrow \infty \text{ as } x \rightarrow 0^+ \end{array}$$

Besides, $\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2} \neq 0$ By L'Hôpital's Rule.

$$\lim_{x \rightarrow 0^+} x \ln x \stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(1/x)}$$

$$= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}$$

Example. Use the L'Hôpital's rule to help sketch the graph of $f(x) = xe^x$.

Sln. Both x, e^x becomes large as $x \rightarrow \infty$ then $\lim_{x \rightarrow \infty} xe^x = \infty$.

As $x \rightarrow -\infty$, but $e^x \rightarrow 0$. So,

$$\lim_{x \rightarrow -\infty} xe^x = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}}$$

$$\stackrel{L'H}{=} \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}} = -\lim_{x \rightarrow -\infty} (e^x) = 0.$$

thus the x -axis is a horizontal asymptote.

$$f'(x) = xe^x + e^x = (x+1)e^x.$$

$\text{Dom } f = \mathbb{R}$. $f(x)$ is increasing for x in $(-1, \infty)$

$f(x)$ is decreasing for x in $(-\infty, -1)$.

$$f''(x) = (x+1)e^x + e^x = (x+2)e^x.$$

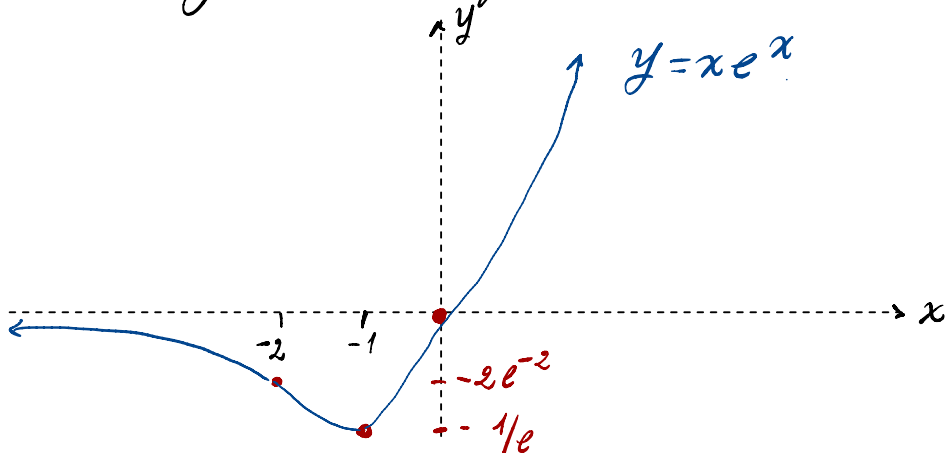
$f(x)$ is $\cup$ on $(-2, \infty)$ and $\cap$ on $(-\infty, -2)$.

$x = -1$ is the only critical point and

$f'(-1) = (-1+2)e^{-1} = e^{-1} > 0$ By the second derivative test at $x = -1$ f has a local minimum.

$f(-1) = -1e^{-1} = -\frac{1}{e}$ $P(-1, -\frac{1}{e})$ is a local minimum.

$x = -2$ is the only inflection point, i.e., $Q(-2, -2e^{-2})$



* Indeterminate Differences.

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$ then the limit

$$\lim_{x \rightarrow a} [f(x) - g(x)]$$

is called an indeterminate form of type $\infty - \infty$.

Example. Compute $\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x)$.

Soln Notice that $\sec x, \tan x \rightarrow \infty$ as $x \rightarrow \frac{\pi}{2}^-$. However,

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x) = \lim_{x \rightarrow \frac{\pi}{2}^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{1 - \sin x}{\cos x}$$

$$\stackrel{L'H}{=} \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{-\cos x}{-\sin x} = 0.$$

* Indeterminate Powers.

Several indeterminate forms arise from the limit

$$\lim_{x \rightarrow a} [f(x)]^{g(x)}.$$

1). $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ type 0^0

2). $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = 0$ type ∞^0

3). $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \infty$ type 1^∞

For each of these three cases we can use

$$y = [f(x)]^{g(x)} \quad \text{then} \quad \ln y = g(x) \ln(f(x))$$

$$y = e^{\ln y} = e^{g(x) \cdot \ln(f(x))}.$$

Example. Calculate $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x}$.

Sln. if $x \rightarrow 0^+$ then $(1 + \sin 4x) \rightarrow 1$ and $\cot x \rightarrow \infty$. then we have 1^∞ .

We know

$$y = e^{\cot x \cdot \ln(1 + \sin 4x)}$$

Since the function e^x is continuous then

$$\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = e^{\lim_{x \rightarrow 0^+} \cot x \cdot \ln(1 + \sin 4x)}$$

$$\lim_{x \rightarrow 0} \cot x \cdot \ln(1 + \sin 4x) = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin 4x)}{1/\cot x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 4x)}{\tan x} \quad \left(\frac{0}{0}\right).$$

$$\begin{aligned} \text{By } L'H &= \lim_{x \rightarrow 0^+} \frac{\frac{4 \cos 4x}{1 + \sin 4x}}{\frac{1}{\cos^2 x}} \\ &= \lim_{x \rightarrow 0^+} \frac{4 \cos^2 x \cos 4x}{1 + \sin 4x} = 4. \end{aligned}$$

$$\text{then } \lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = e^4$$

Example. Find $\lim_{x \rightarrow 0^+} x^x$

Sln.

$$x^x = e^{\ln(x^x)} = e^{x \ln x}$$

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \ln x} = e^{\lim_{x \rightarrow 0^+} x \ln x}$$

$$\lim_{x \rightarrow 0^+} x \ln x = 0. \quad (\text{By previous example}).$$

$$\text{then} \quad \lim_{x \rightarrow 0^+} x^x = e^0 = 1.$$